

PURNA MUMMANI

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Education

University of North Carolina at Charlotte GPA: 3.8/4

Master of Science in Computer Science (Concentration in Data Science)

Jan 2024 – Dec 2025

Charlotte, NC

Vellore Institute of Technology, India

Bachelor of Technology in Computer Science and Engineering

Aug 2019 – July 2023

India

Experience

UNC Charlotte: OneIT

Student IT Support Technician

Aug 2024 – May 2025

Charlotte, NC

- * Delivered frontline IT support by resolving **15–20 weekly technical issues** via the **Mercury ticketing system**, troubleshooting Wi-Fi, hardware, and software for Windows/macOS devices; facilitated campus-wide move-in technology setups by onboarding resident devices and coordinating with senior IT staff to escalate complex incidents while ensuring clear end-user communication and 100% incident resolution.

Cognizant

Software Engineer

Jan 2023 – Dec 2023

Hyderabad, India

- * Optimized **React.js** UI responsiveness and component reusability to **speed up page loads by 28%**, while architecting **Next.js SSR/REST APIs** to drive a **22% increase** in organic traffic.
- * Migrated legacy features to **Spring Boot/Docker** microservices to reduce service errors by **24%** and integrated **Redis** caching to decrease API response times by **18%** for high-concurrency handling.
- * Enforced robust security protocols, including **JWT, RBAC, and AES encryption**, to protect sensitive data for **50+ internal users** while maintaining strict organizational and data compliance.

Cognizant

Software Engineer Intern

July 2022 – Jan 2023

Hyderabad, India

- * Developed backend features using **Java/Spring Boot** and executed **100+ tests** to cut bugs by **18%** while reducing requirement gaps by **12%** via **Postman API testing**.
- * Increased code coverage by **25%** using **JUnit/TestNG** and managed **CI/CD pipelines** to ensure seamless production releases with zero downtime and strengthened reliability.

Technical Skills

Programming: Java, Python, C# , C, C++, HTML/CSS, JavaScript, SQL

Frameworks/Softwares: Angular, .NET Core, ASP.NET, ReactJS, NodeJS, Spring Boot, W3af, Metasploit, Jira, WordPress, Apache spark, Hadoop, Kafka

Database & Cloud: MySQL, PostgreSQL, NoSQL (MongoDB), Entity Framework, MS Azure, AWS, Docker, Kubernetes, CloudFormation, API Gateway

Developer Tools: VS, Eclipse, Virtual Box, VM ware, Git, MSSQL, Swagger, Figma, Jupyter, Google Colab, Cisco, Tableau, PowerBI

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, OpenCV

other Tools: Data Bricks, Social Engineering Toolkit, Burp Suite, Nmap, Sqlmap, Wireshark, SNORT, SQL workbench **OS:** Windows, MAC, Linux, Kali Linux, Ubuntu

Projects

Netflix-GPT | React, OpenAI, Firebase, TailwindCSS

Nov 2025

- Engineered a high-performance movie streaming platform that lowered bounce rates through seamless navigation and Incorporated the TMDB API to boost user engagement by 35% with real-time content; further Leveraged **OpenAI** for AI-powered search, enhancing recommendations and accuracy, while boosted system performance to decrease load times by 25% through a mobile-responsive design.

Hybrid Transformer-CNN for Dense Video Analysis | PyTorch, Python, Transformers, OpenCV

Sep 2025

- Designed and implemented a full-scale MS-TCT (ConvTransformer) in PyTorch, utilizing a 4-stage temporal encoder to capture atomic and composite actions in untrimmed videos, achieving a **+9.8% mAP boost** over I3D baselines.
- Customized a Gaussian Heat-map branch and Temporal Scale Mixer to unify multi-resolution features, optimizing the system to use **85% fewer FLOPs** while maintaining real-time frame-level detection at an **8 FPS sampling rate**.

Career War Room: AI Resume Insight & Job Matcher | Python, Ollama, NLP, Streamlit

May 2025

- Developed a privacy-focused intelligence platform by migrating to **Local LLMs (Ollama)** and **Sentence Transformers** for semantic similarity to rank job listings, and deployed a full-stack Streamlit dashboard that automates skill-gap analysis via an **asynchronous FastAPI**.

Doctor-On-Call: Healthcare Management System | JavaScript, MongoDB, Express, React, Node.js

Aug 2024

- Architected a multi-tiered healthcare portal featuring secure **JWT-based authentication**, multi-role authorization, and an automated doctor-onboarding pipeline and real-time status notifications, hosted on **AWS** and refined scheduling engine, reducing post-release logic issues by ~40% while securing sensitive patient data and ensure high reliability during peak traffic.

US Aviation Intelligence: Market & Fare Dynamics Analysis | Tableau, Dashboard, Visual Analytics

May 2024

- Built a Tableau Story for US aviation trends analyzing 10,000+ flight records, utilizing K-means clustering and geospatial mapping to identify carrier dominance and uncover pricing inefficiencies.

Achievements & Certifications

Awarded for Technical Excellence at Engineering Clinics Expo 2021 for developing a real-time, Arduino-based robotic hand system from a pool of 250+ competing teams. Microsoft Certified: Azure Fundamentals (AZ-900, AWS Academy Graduate, Emotion AI Certification : Facial Key-points Detection.